THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT 1

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1. Write a program (WAP) to display "Hello World" on console display.

CODE:

```
#include <iostream>
using namespace std;
int main(){
   cout<<"Hello World" <<endl;
return 0;}
```

```
Hello World
```

2. WAP to that reads two values from keyboard and displays the larger value on the screen.

CODE:

```
#include <iostream>
using namespace std;
int main() {
   int num1,num2;
   cin>>num1>>num2;
(num1>num2) ? (cout<<"Larger value is: "<<num1) : ((num2=num1) ? (cout<<"Both are equal."):(cout<<"Larger value is: "<<num2));
return 0;}
```

```
50
100
Larger value is: 100
```

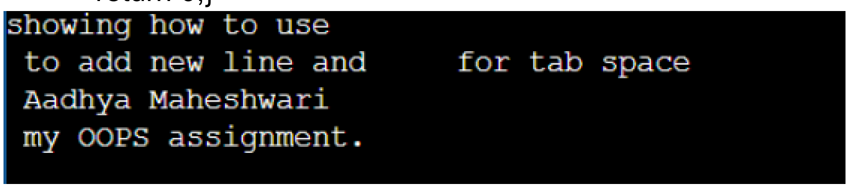
3. WAP to read three values a, b, c from keyboard and display the value of x = a/b-c.

CODE:

```
#include <iostream>
using namespace std;
int main() {
   float a,b,c,x;
   cin>>a>>b>>c;
   x = (a/b-c);
   cout<<”x equals: ”<<x;
   return 0;}
```

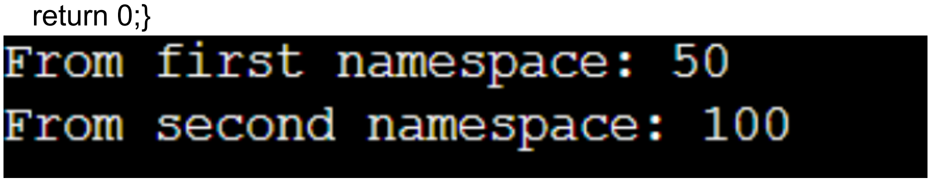
```
5 7 9
x equals: -8.28571
```

4. WAP to implement the following control characters:
- ‘\n’ is for new line, or you can use endl – cout<<endl<<“message”;
- ‘\t’ is for tab
- ‘\a’ is an alarm sound
- ‘\r’ is carriage return to go to the beginning of the current line

CODE:
```
#include <iostream>
using namespace std;
int main() {
        cout<<"showing how to use \n to add new line and \t for tab space";
        cout<<"\n Aadhya Maheshwari \a"<<endl;
        cout<<"I am completing my OOPS \r assignment.";
        return 0;}
```


```
showing how to use
 to add new line and      for tab space
 Aadhya Maheshwari
 my OOPS assignment.
```

5. Implement namespace in a program to illustrate the use of same name
 variables and functions in different sections/libraries of the code

CODE:
```
#include <iostream>
using namespace std;
namespace one{
   int value =50;}
namespace two{
   int value=100;}
int main(){
   cout<<"From first namespace: "<<one::value<<endl;
   cout<<"From second namespace: "<<two::value<<endl;
   return 0;}
```


```
From first namespace: 50
From second namespace: 100
```

6. Write a program to define a structure Student that contains 3 variables and 2 functions: - Variables: (a) name (b) roll (c) marks - Functions: (1) setStudentData() (2) getStudentData() Use loops to input and output data for 3 student object entries.

CODE:
```
#include <iostream>
using namespace std;
struct student{
   string  name;
   int roll, marks;
void setstudent(){
   cout << "Enter name: ";
   cin >> name;
```

```
    cout << "Enter roll number: ";
    cin >> roll;
    cout << "Enter marks: ";
    cin >> marks;}
void getstudent(){
    cout << "\nName: " << name;
    cout << "\nRoll Number: " << roll;
    cout << "\nMarks: " << marks << endl;}};
int main(){
    student s[3];
    for (int i = 0; i < 3; i++){
        cout << "\nEnter details of student " << i + 1 << endl;
        s[i].setstudent();}
    for (int i = 0; i < 3; i++){
        cout << "\nDetails of student " << i + 1 << endl;
        s[i].getstudent();}
    return 0;}
```

```
Enter details of student 1
Enter name: Aadhya
Enter roll number: 212
Enter marks: 100

Enter details of student 2
Enter name: Hiyaanshi
Enter roll number: 142
Enter marks: 99

Enter details of student 3
Enter name: Bhavya
Enter roll number: 418
Enter marks: 100

Details of student 1

Name: Aadhya
Roll Number: 212
Marks: 100

Details of student 2

Name: Hiyaanshi
Roll Number: 142
Marks: 99

Details of student 3

Name: Bhavya
Roll Number: 418
Marks: 100
```

7. In the previous program make the data private, and functions public.

CODE:

```
#include <iostream>
using namespace std;
class student{
private:
    char name[50];
    int roll;
    float marks;
public:
    void setstudent(){
        cout << "Enter name: ";
        cin >> name;
        cout << "Enter roll number: ";
        cin >> roll;
        cout << "Enter marks: ";
        cin >> marks;}
    void getstudent(){
        cout << "\nName: " << name;
        cout << "\nRoll Number: " << roll;
        cout << "\nMarks: " << marks << endl;};

int main()
{student s[3];
    for (int i = 0; i < 3; i++){
        cout << "\nEnter details of student " << i + 1 << endl;
        s[i].setstudent();}
    for (int i = 0; i < 3; i++){
        cout << "\nDetails of student " << i + 1 << endl;
        s[i].getstudent();}
    return 0;}
```

```
Enter details of student 1
Enter name: Aadhya
Enter roll number: 212
Enter marks: 100

Enter details of student 2
Enter name: Hiyaanshi
Enter roll number: 142
Enter marks: 99

Enter details of student 3
Enter name: Bhavya
Enter roll number: 418
Enter marks: 100

Details of student 1

Name: Aadhya
Roll Number: 212
Marks: 100

Details of student 2

Name: Hiyaanshi
Roll Number: 142
Marks: 99

Details of student 3

Name: Bhavya
Roll Number: 418
Marks: 100
```

8. Convert program #4 from struct to class.

CODE:

```
#include <iostream>
using namespace std;
class student {
private:
   string name;
   int roll;
   int marks;
public:
   void setstudent() {
      cout << "Enter name: ";
      cin >> name;
```

```
        cout << "Enter roll number: ";
        cin >> roll;
        cout << "Enter marks: ";
        cin >> marks;}
    void getstudent() {
        cout << "\nName: " << name;
        cout << "\nRoll Number: " << roll;
        cout << "\nMarks: " << marks << endl;}};

int main() {
    student s[3];
    for (int i = 0; i < 3; i++) {
        cout << "\nEnter details of student " << i + 1 << endl;
        s[i].setstudent();}
    for (int i = 0; i < 3; i++) {
        cout << "\nDetails of student " << i + 1 << endl;
        s[i].getstudent();}
    return 0;}
```

```
Enter details of student 1
Enter name: Aadhya
Enter roll number: 212
Enter marks: 100

Enter details of student 2
Enter name: Hiyaanshi
Enter roll number: 142
Enter marks: 99

Enter details of student 3
Enter name: Bhavya
Enter roll number: 412
Enter marks: 100

Details of student 1

Name: Aadhya
Roll Number: 212
Marks: 100

Details of student 2

Name: Hiyaanshi
Roll Number: 142
Marks: 99

Details of student 3

Name: Bhavya
Roll Number: 412
Marks: 100
```

9. Discuss the differences of structures in C and structures in C++ during the lab with other students or your instructor.

10. WAP to read an integer value from the keyboard and display on screen “WELL DONE” t

CODE:

```
#include <iostream>
using namespace std;
int main() {
   int n;
   cout<<"Enter n: ";
   cin>>n;
   for(int i=0; i<n; i++)
      cout<<"WELL DONE"<<endl;
   return 0;}
```